

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/F_DEB_AWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3313, C2 = 6664, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (kapI,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (K,           Uniform,                0.00001,   2);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? TRUE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.00686584	222.05200	0.1034090	0.0135851	0.62997300	0.1219860
2.5%	0.00551884	195.94295	0.1018320	0.0118318	0.00819366	0.1081968
97.5%	0.11669445	247.53800	1.2146600	0.2298820	0.73491200	0.6343279
Min	0.00332135	90.62130	0.1000120	0.0105772	0.00619815	0.1000550
Max	0.27050000	262.06100	1.2469100	0.2674630	0.98934100	0.7802450
Result_mean	0.04111682	221.20093	0.7975702	0.1391230	0.22290414	0.4123143
Result_sd	0.03042338	13.50011	0.4778541	0.0900014	0.29902162	0.1971879

	kapI.1.	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.
Result_mode	0.8990640	77.31520	26.806700	0.0005241570	0.05704390	4.418700
2.5%	0.1000710	45.92790	8.171560	0.0003624990	0.00592200	2.170509
97.5%	0.8998990	128.36800	29.443660	0.0103042000	1.65687100	6.083690
Min	0.1000000	40.04440	6.839840	0.0000748389	0.00419069	1.570310
Max	0.8999990	131.22900	35.239600	0.0491326000	1.99911000	7.457110
Result_mean	0.3674961	103.14207	15.229579	0.0025466703	0.14921192	3.307982
Result_sd	0.3756812	26.56853	7.163761	0.0026989426	0.38969247	1.319889

	wE.1.	Sigma_W.1.
Result_mode	1.069910e-09	0.5206280
2.5%	1.076840e-09	0.4863886
97.5%	6.142110e-07	1.0355100
Min	1.002470e-09	0.4417890
Max	1.134500e-06	4.9332300
Result_mean	8.845227e-08	0.8041956
Result_sd	1.603118e-07	0.2040287

Number of observations, n = 358

Number of parameters, k = 13

Log-likelihood, LL = -482.7218

BIC = 1041.891

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	6.11	31.77
E_coc.1.	6.46	17.77
Ehp.1.	4.48	24.32
Fm.1.	1.11	1.27
K.1.	1.95	9.49
kap.1.	14.40	87.57

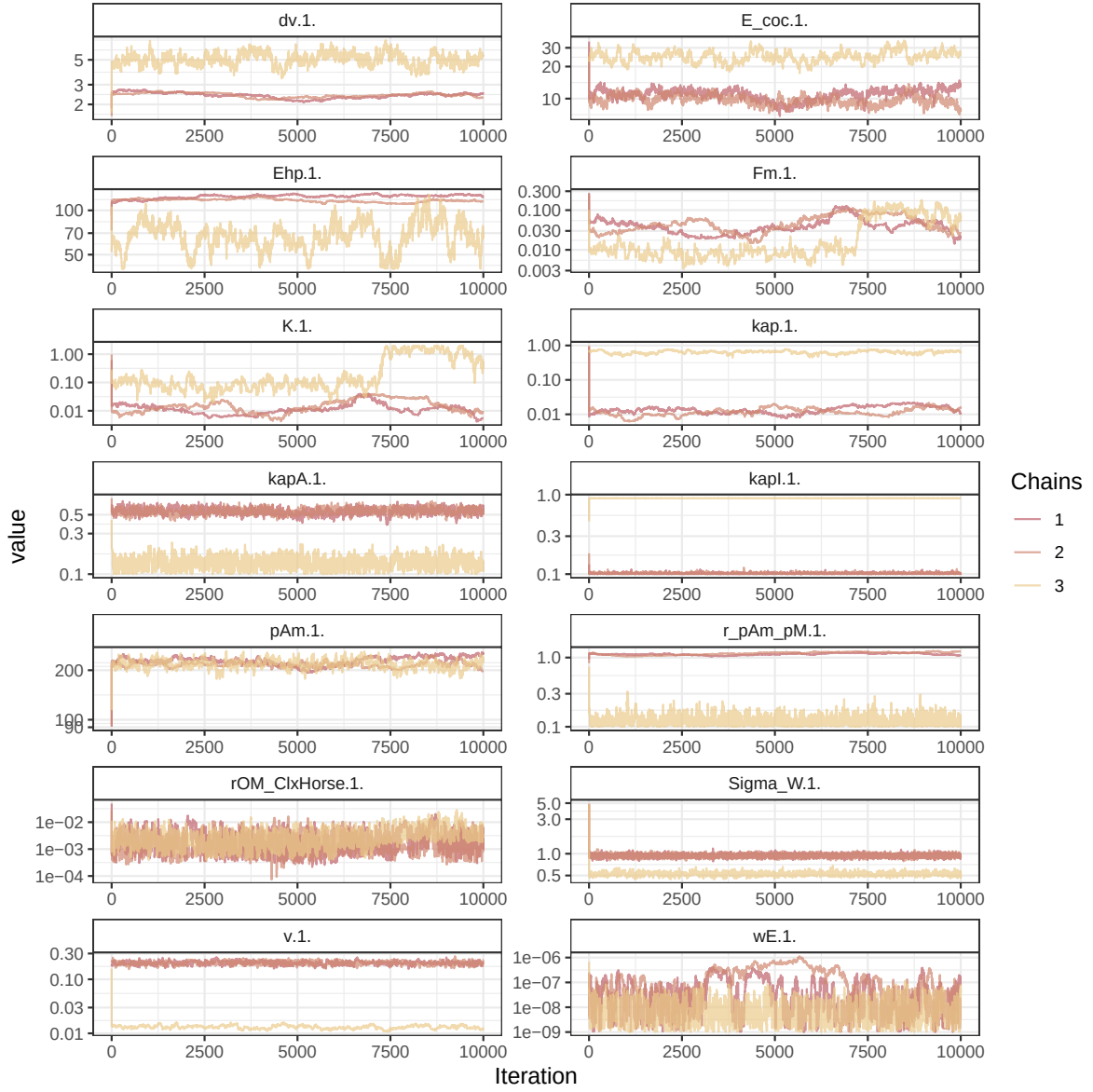


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

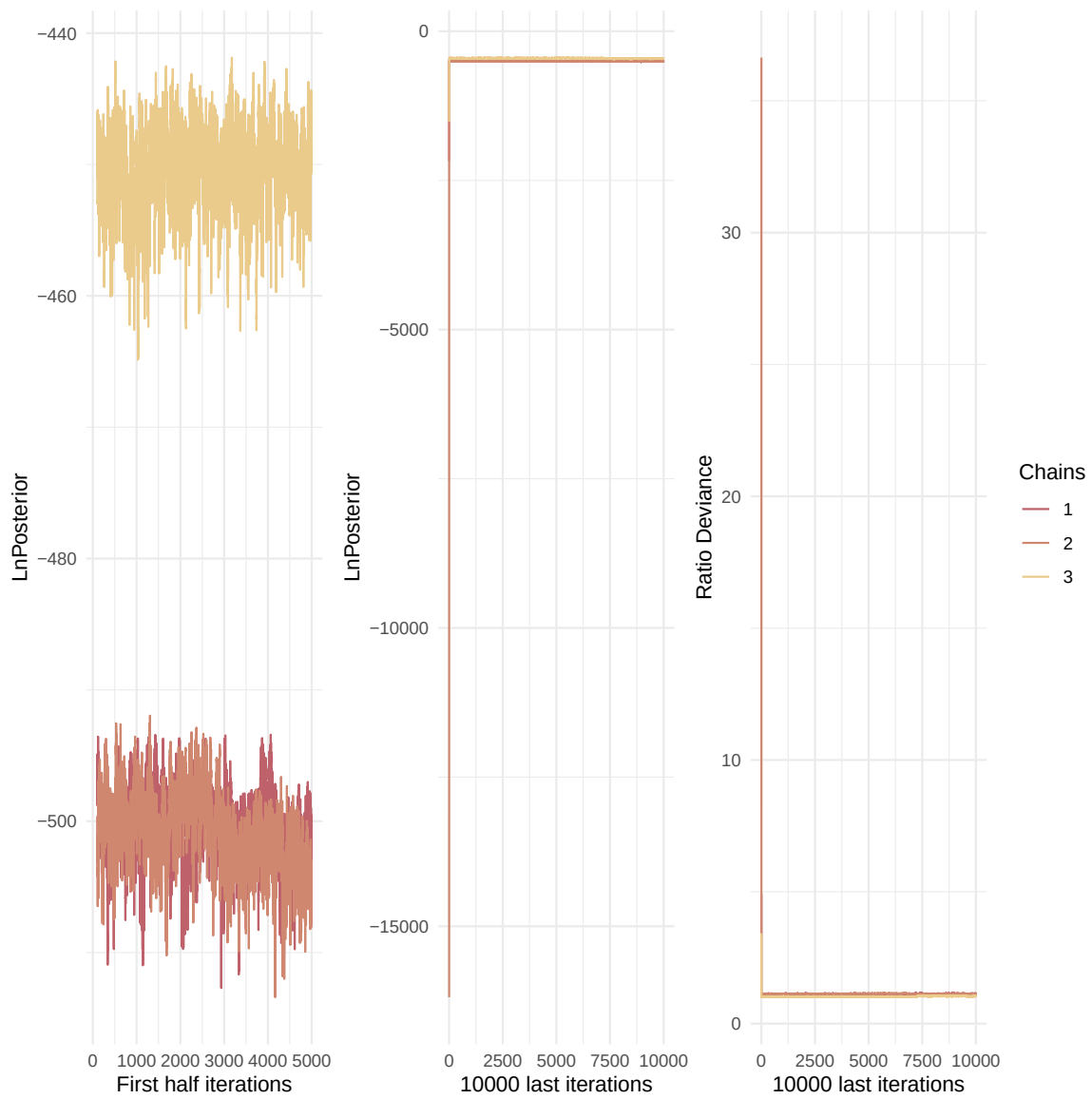


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

kapA.1.	8.66	18.87
kapI.1.	407.93	824.92
pAm.1.	1.30	1.85
r_pAm_pM.1.	39.39	77.98
rOM_ClxFHorse.1.	1.16	1.48
Sigma_W.1.	8.25	16.75
v.1.	13.18	32.25
wE.1.	1.59	5.49

Multivariate psrf

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```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

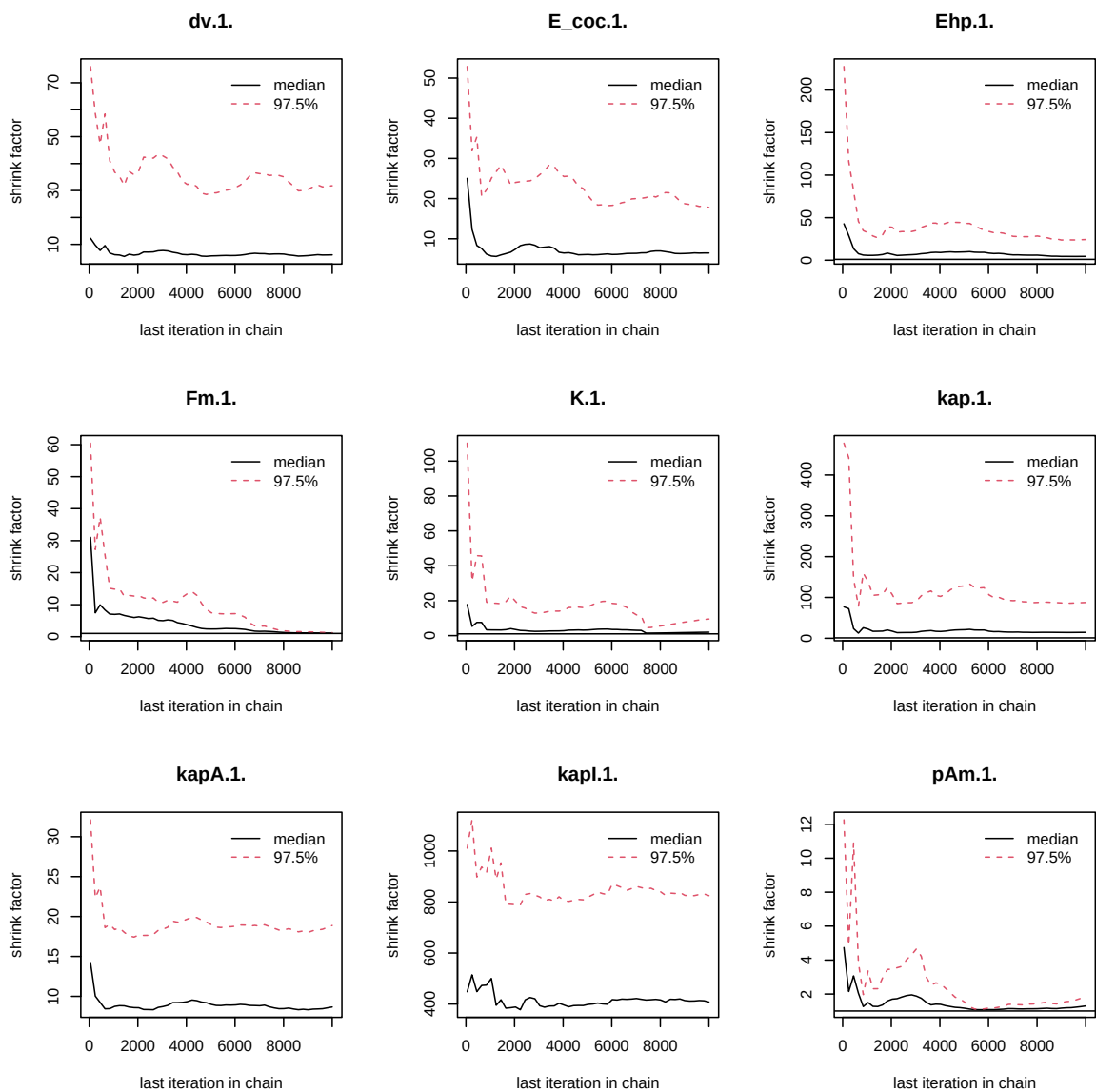


Figure 3: Chains convergence.

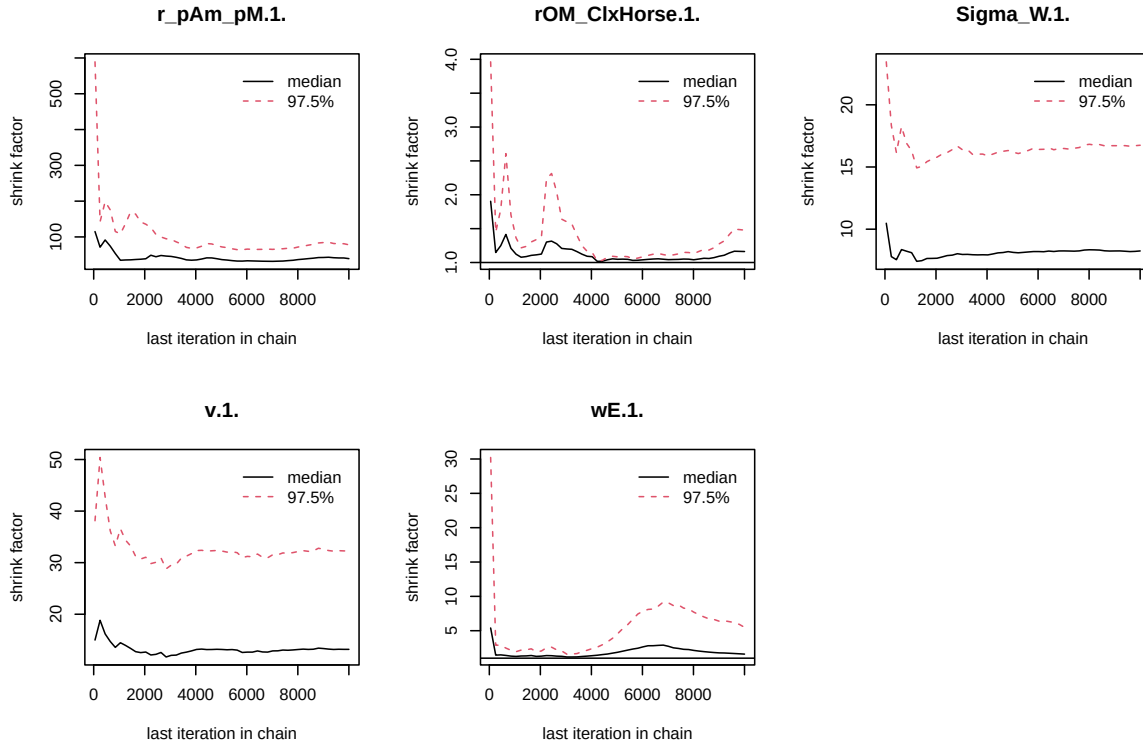


Figure 4: Chains convergence.

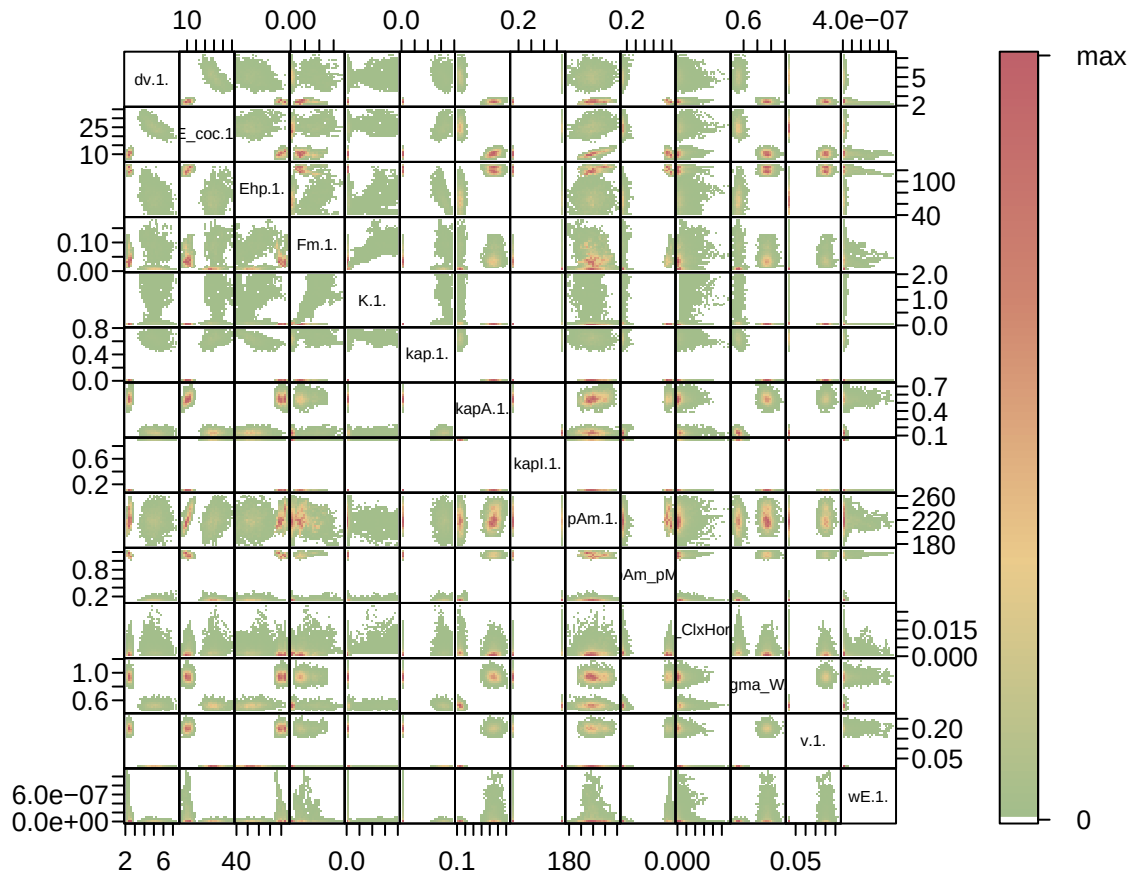


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

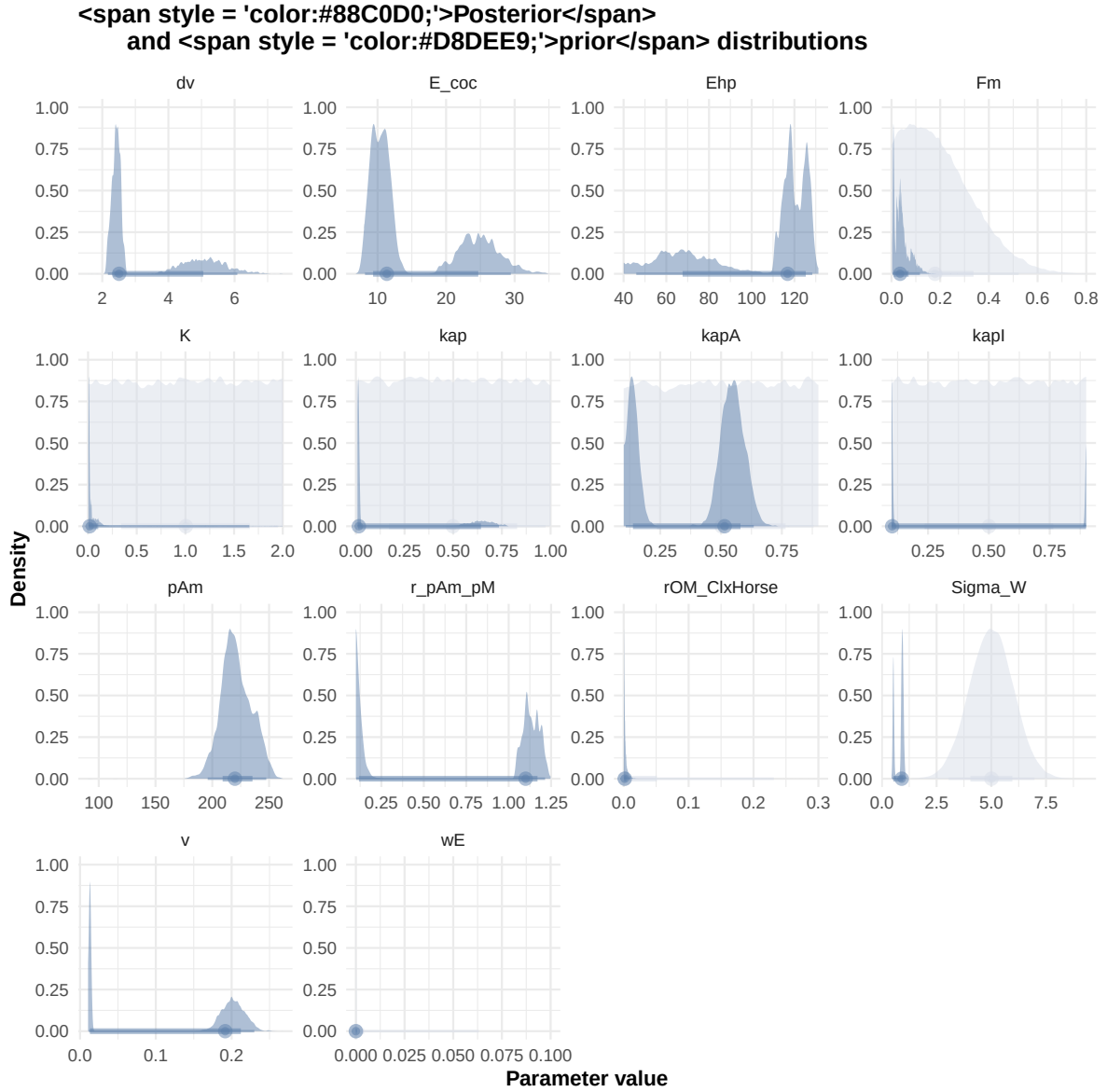


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

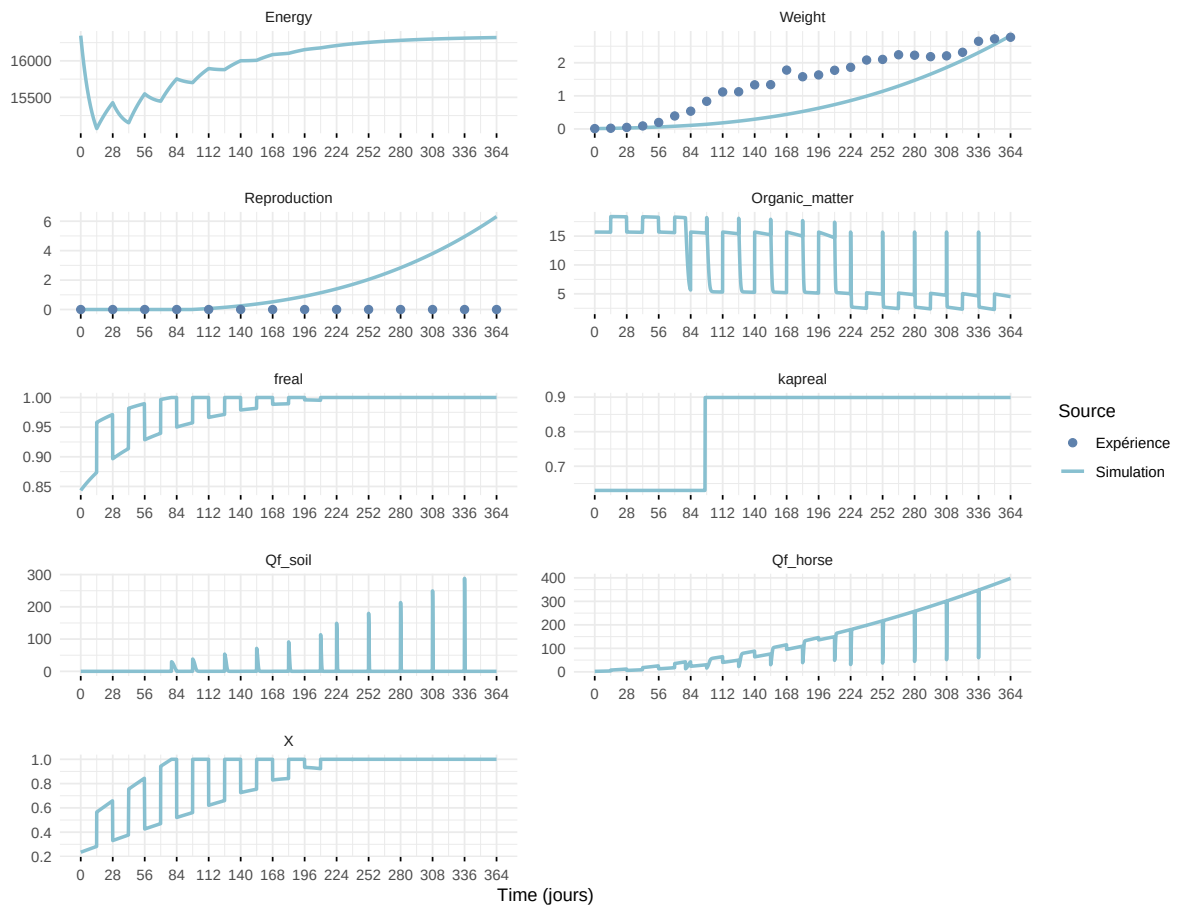
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



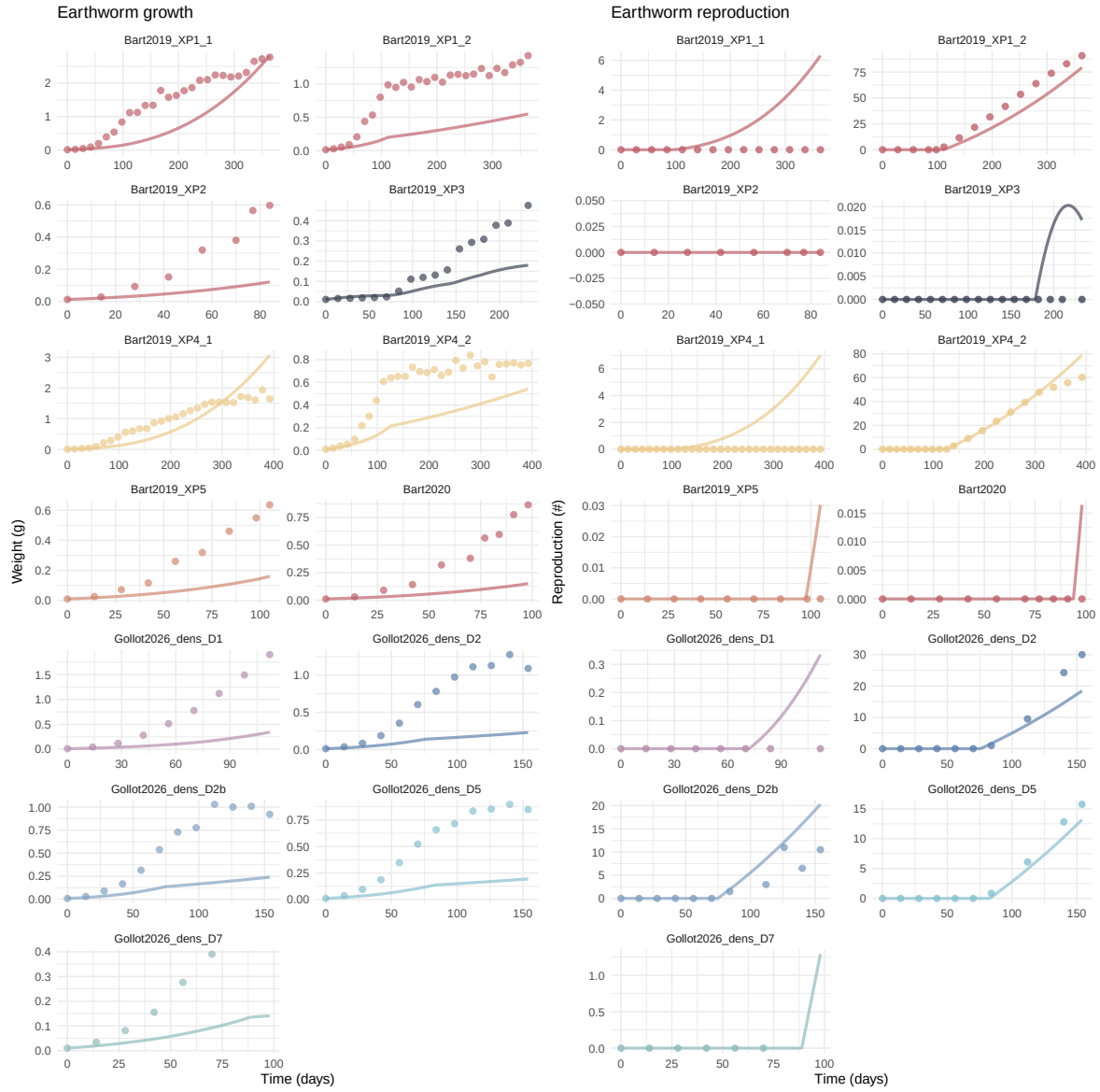
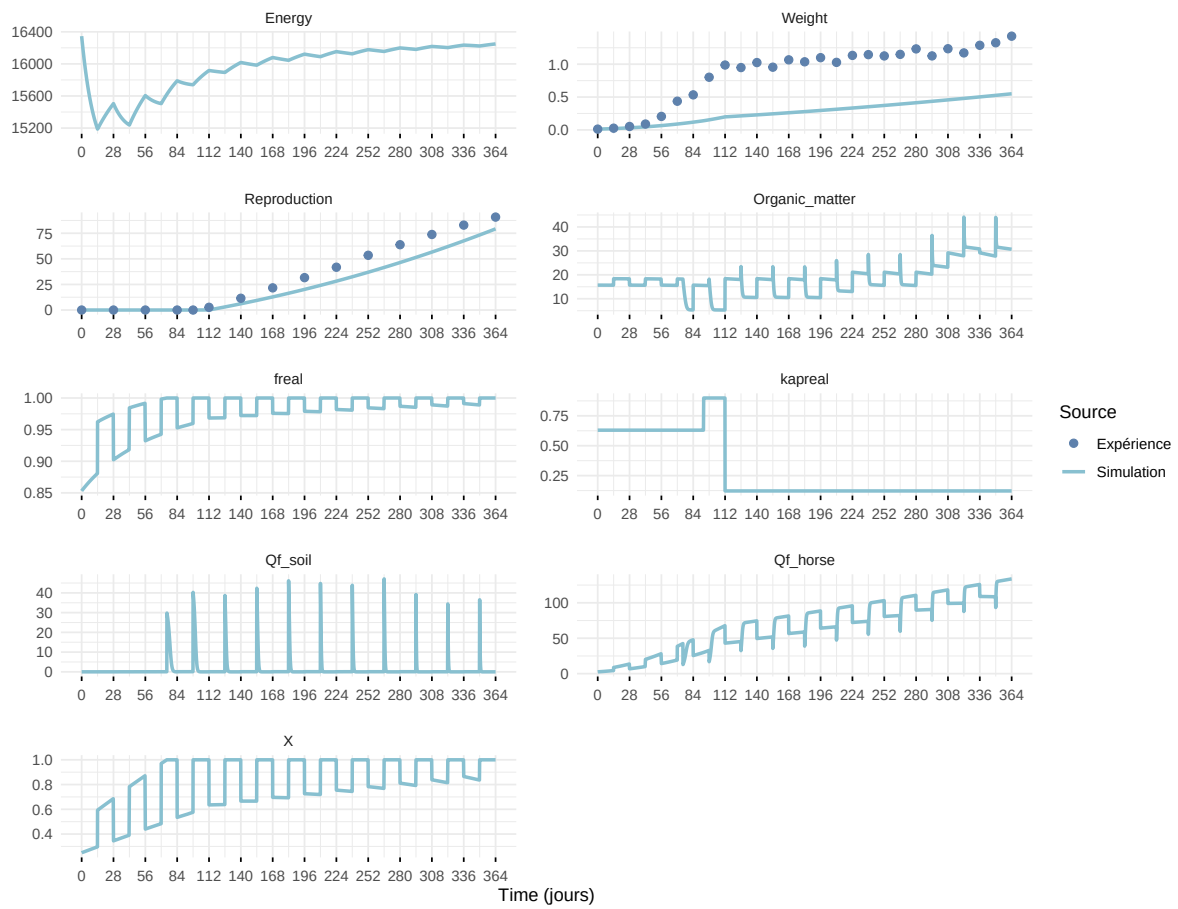
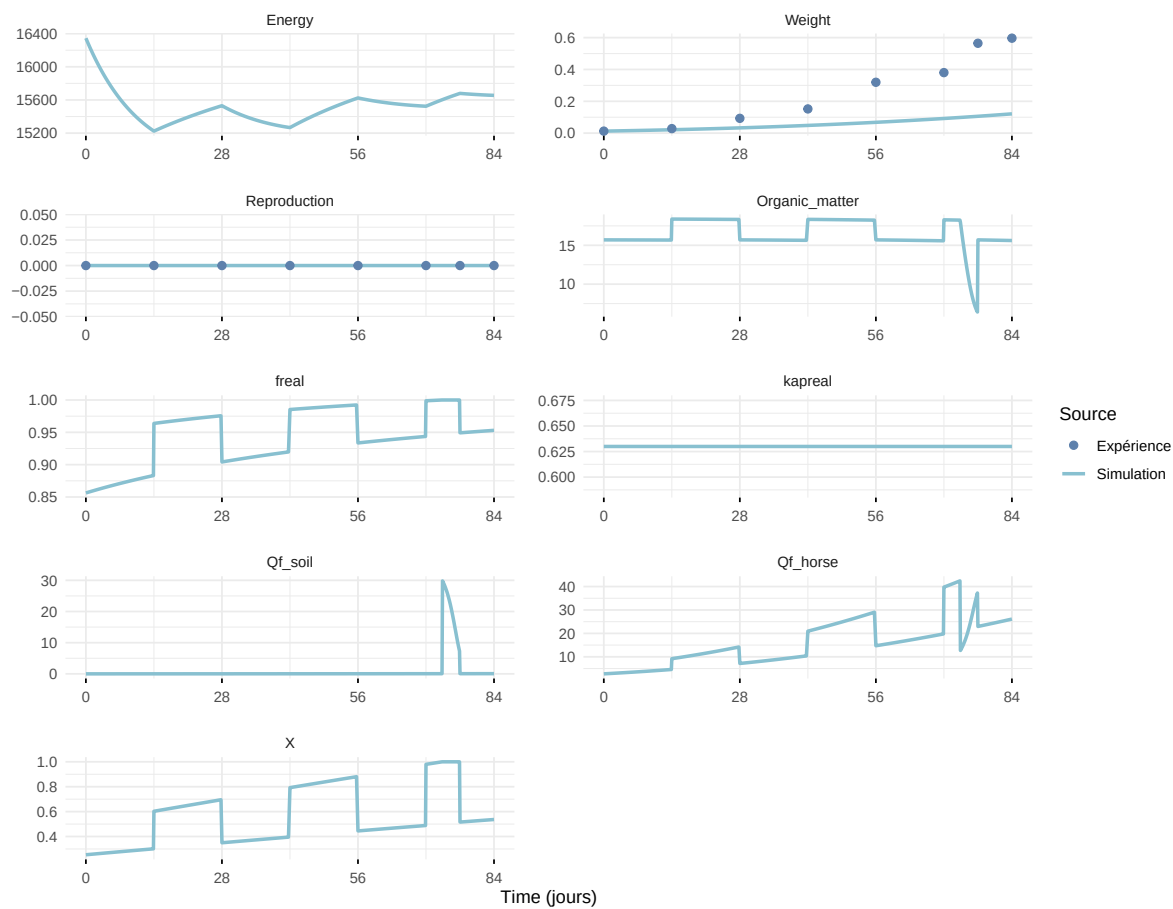


Figure 7: Simulations Weight and Reproduction.

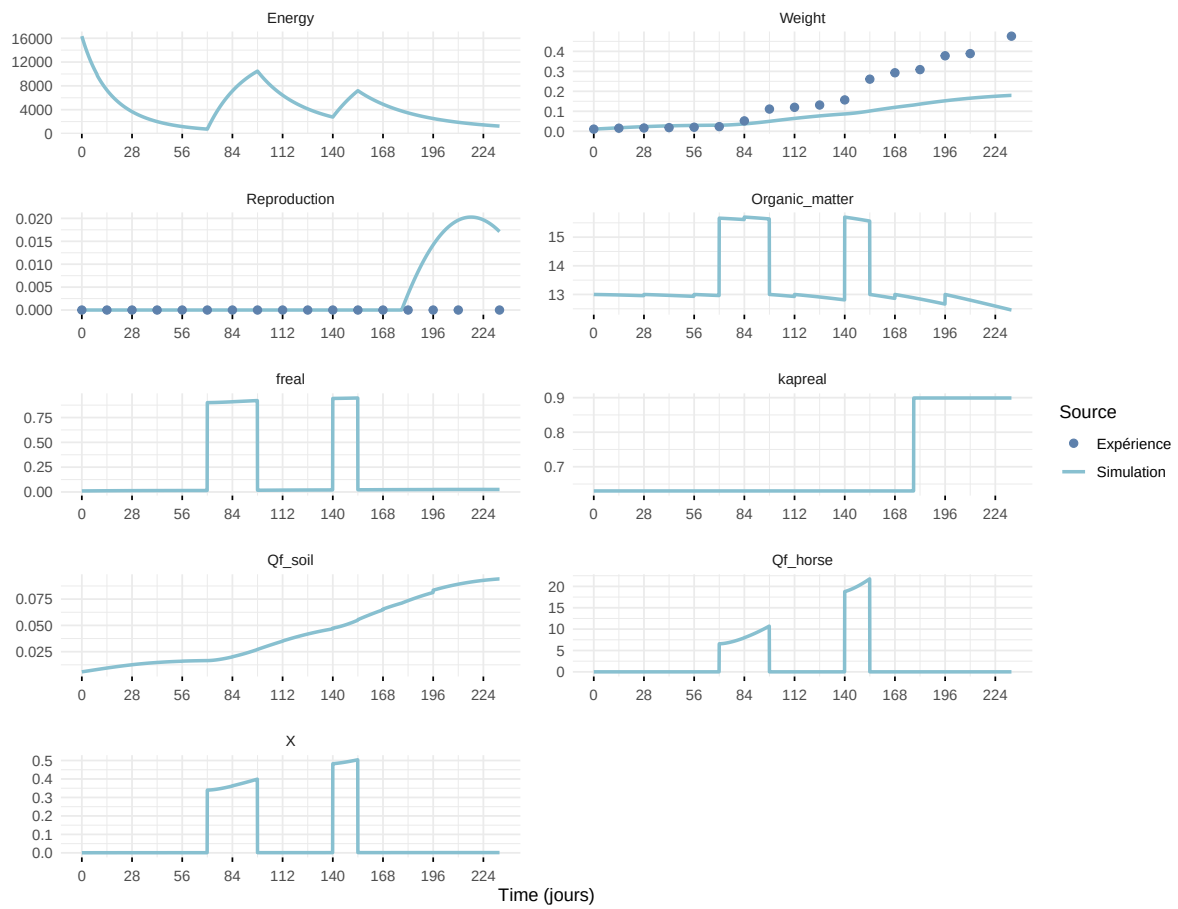
Expérience : Bart2019_XP1_2



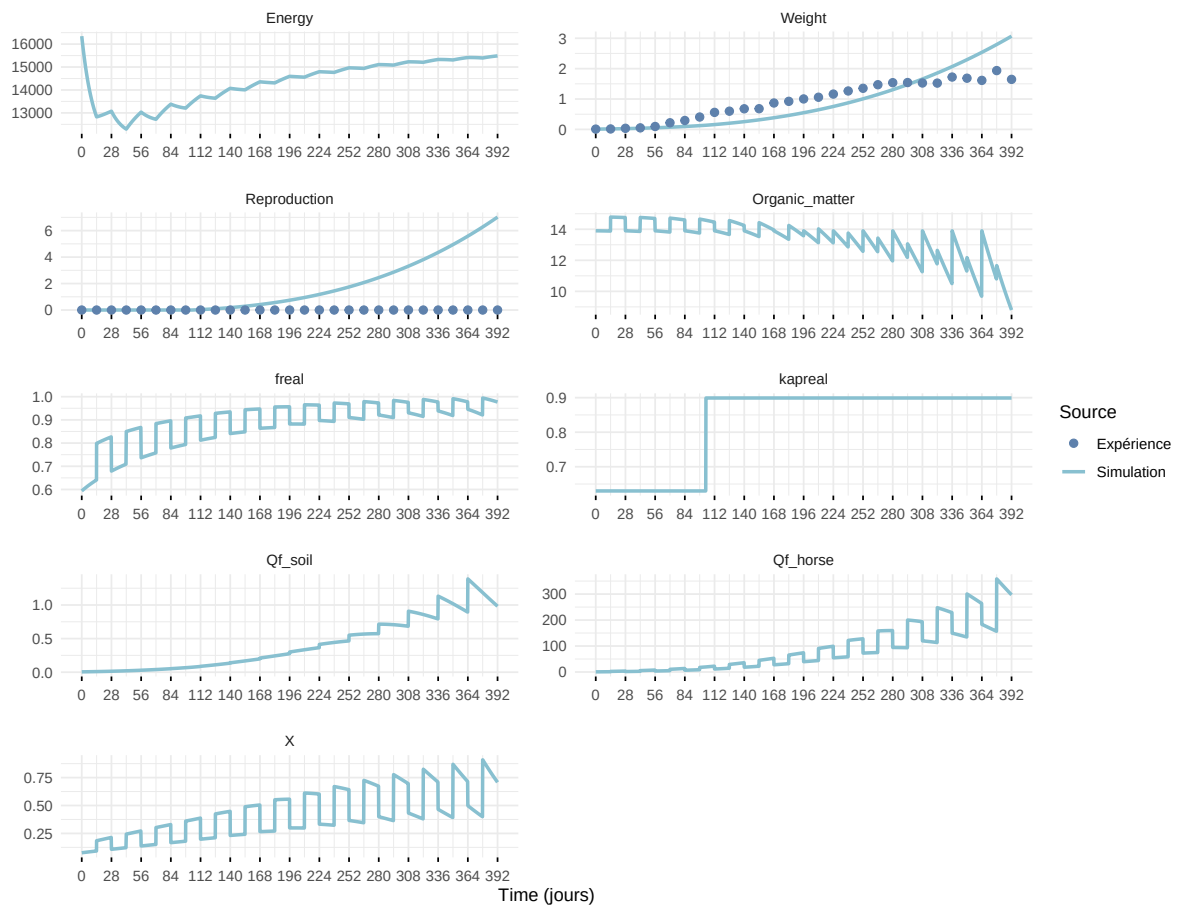
Expérience : Bart2019_XP2



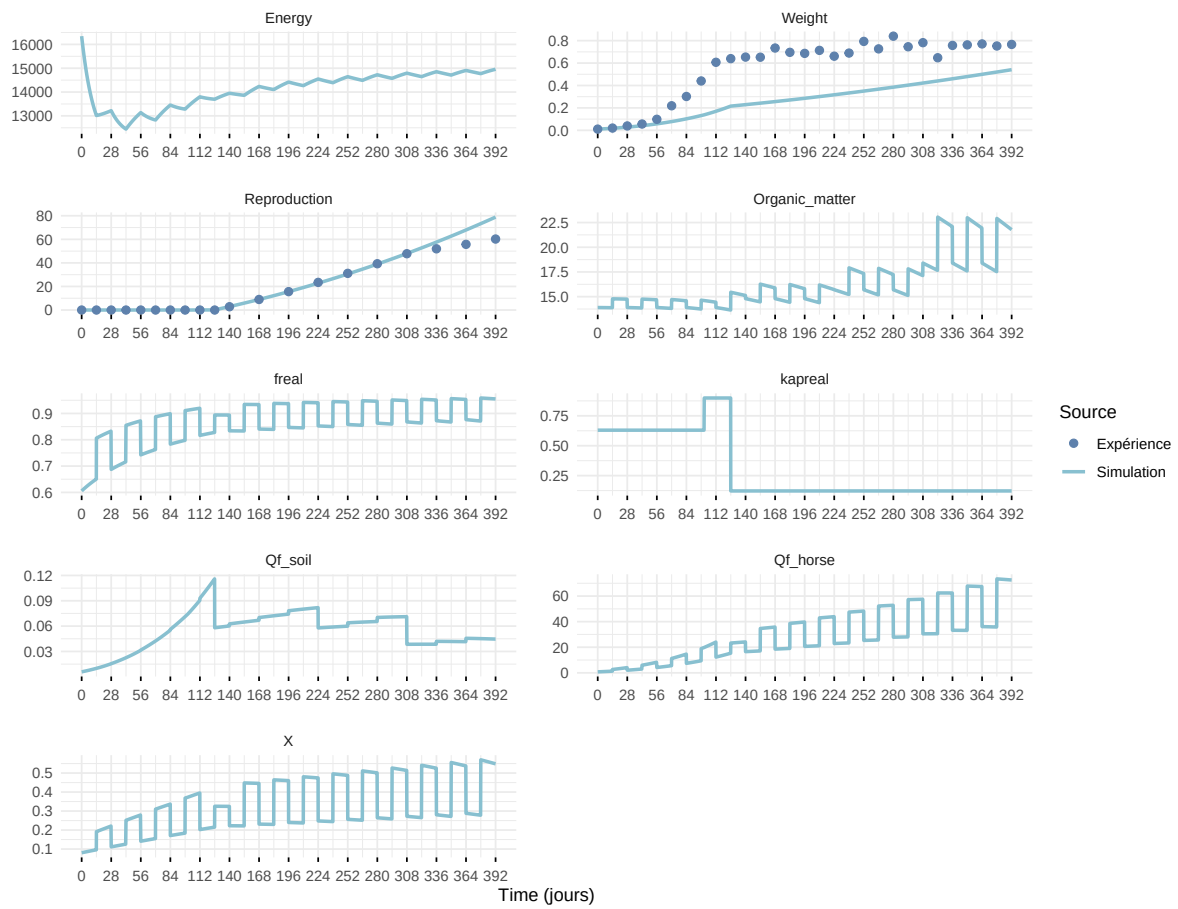
Expérience : Bart2019_XP3



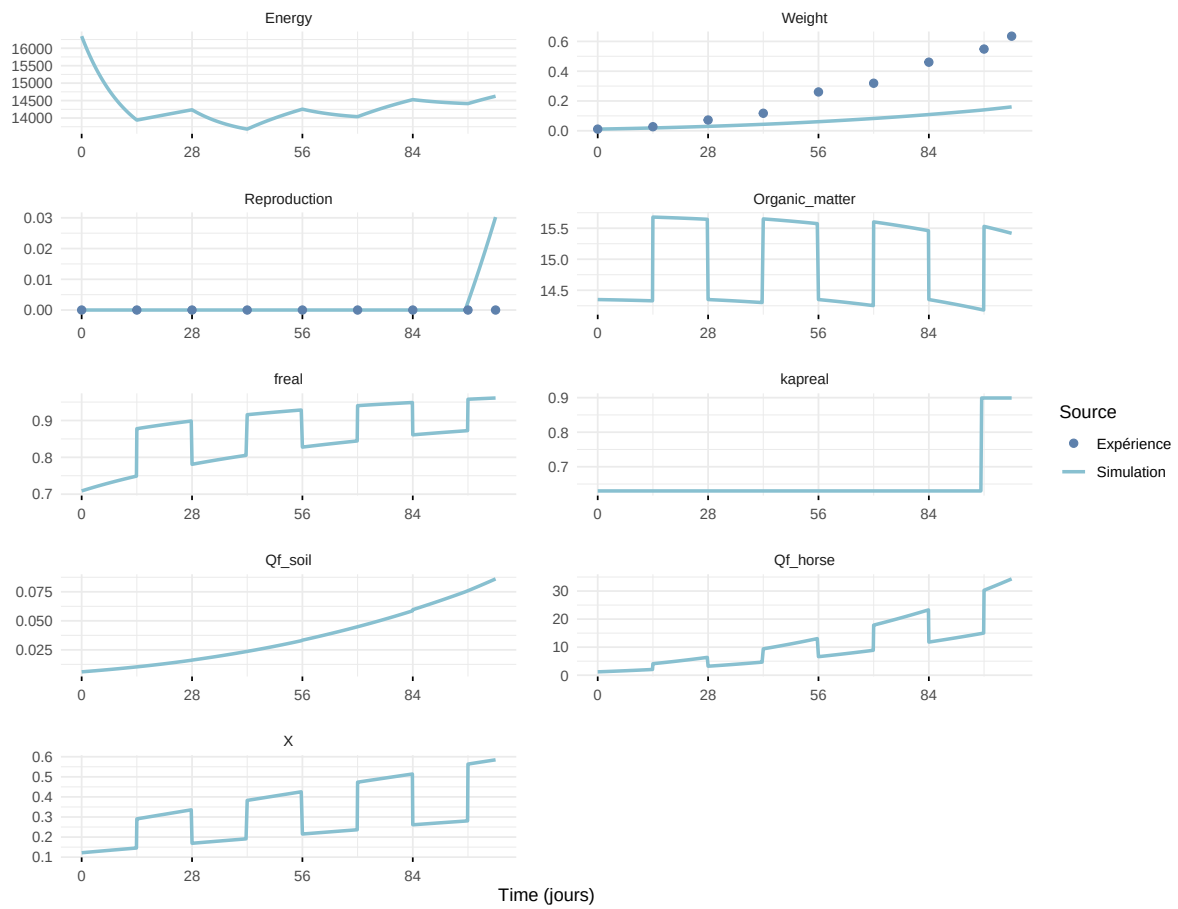
Expérience : Bart2019_XP4_1



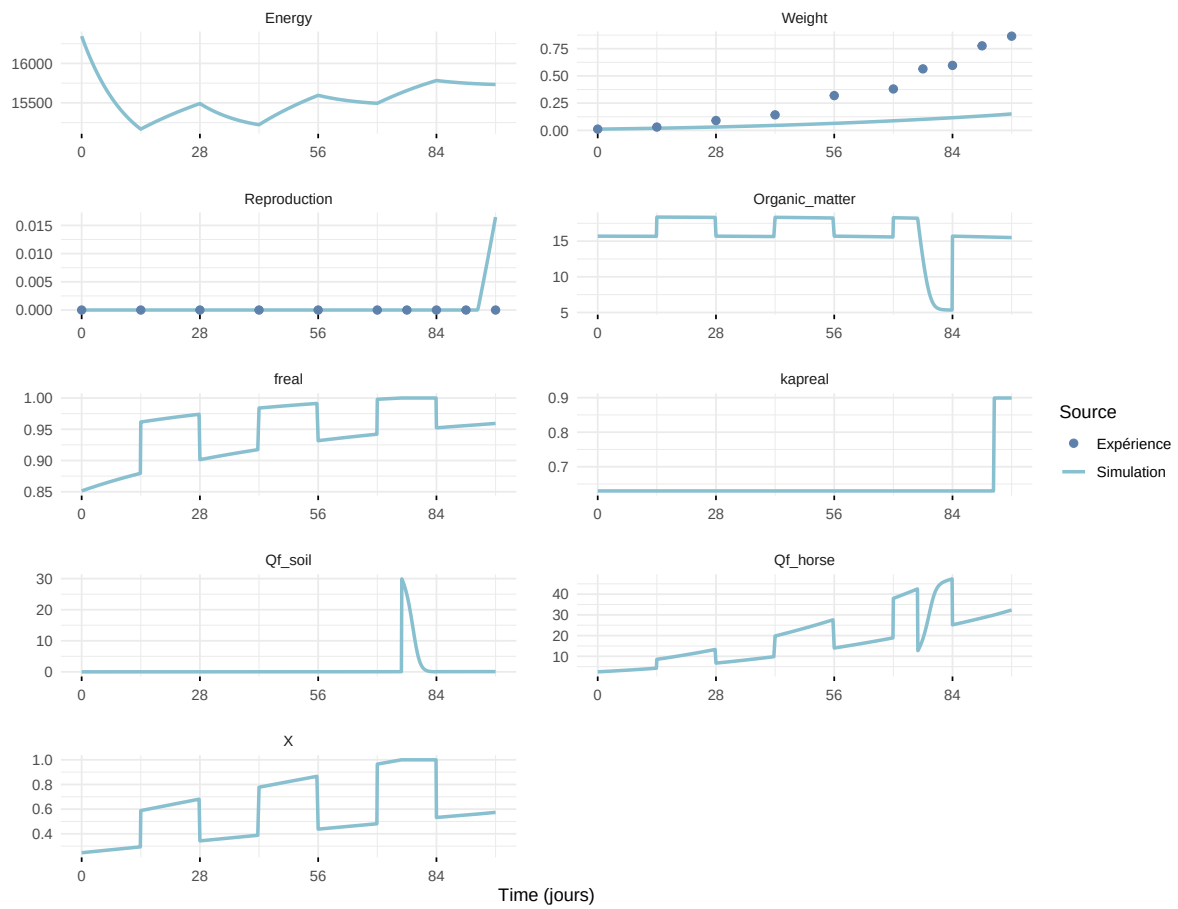
Expérience : Bart2019_XP4_2



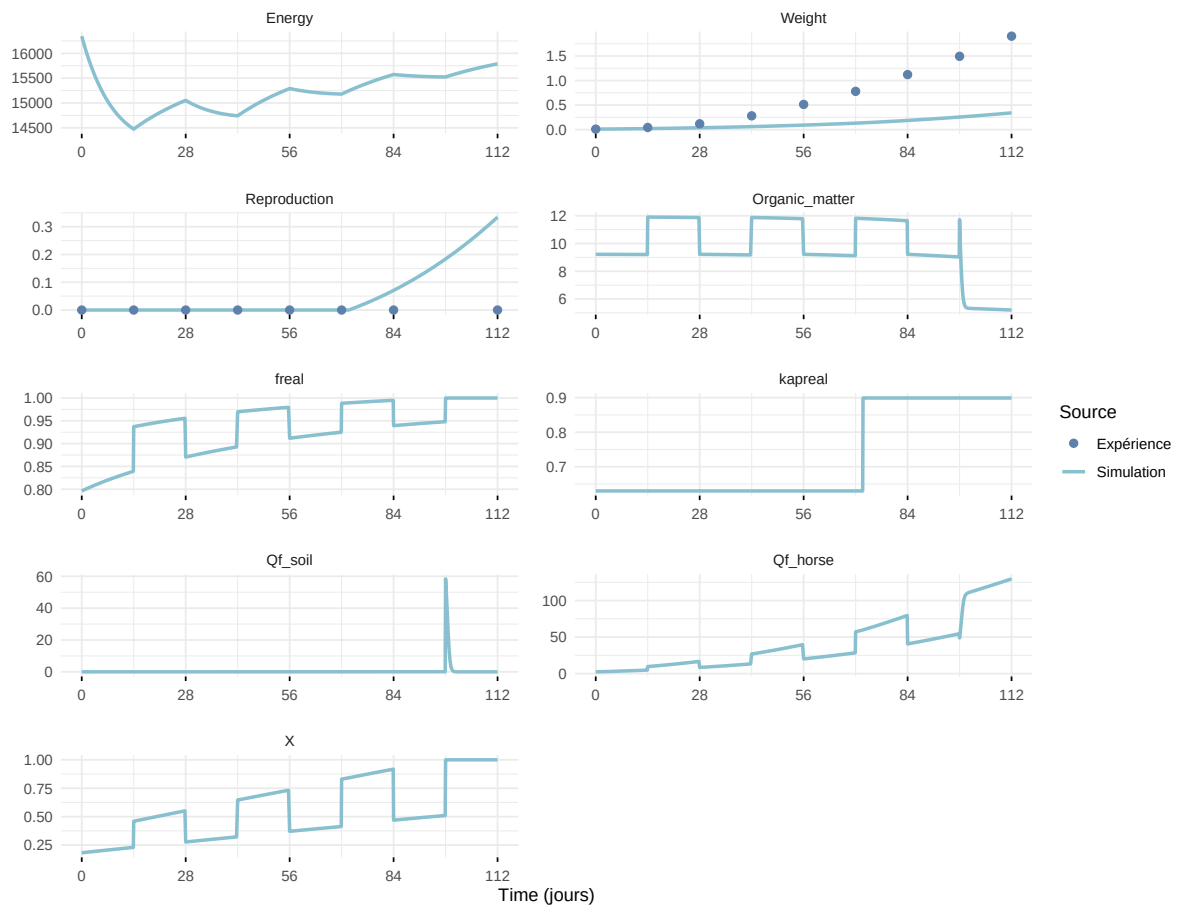
Expérience : Bart2019_XP5



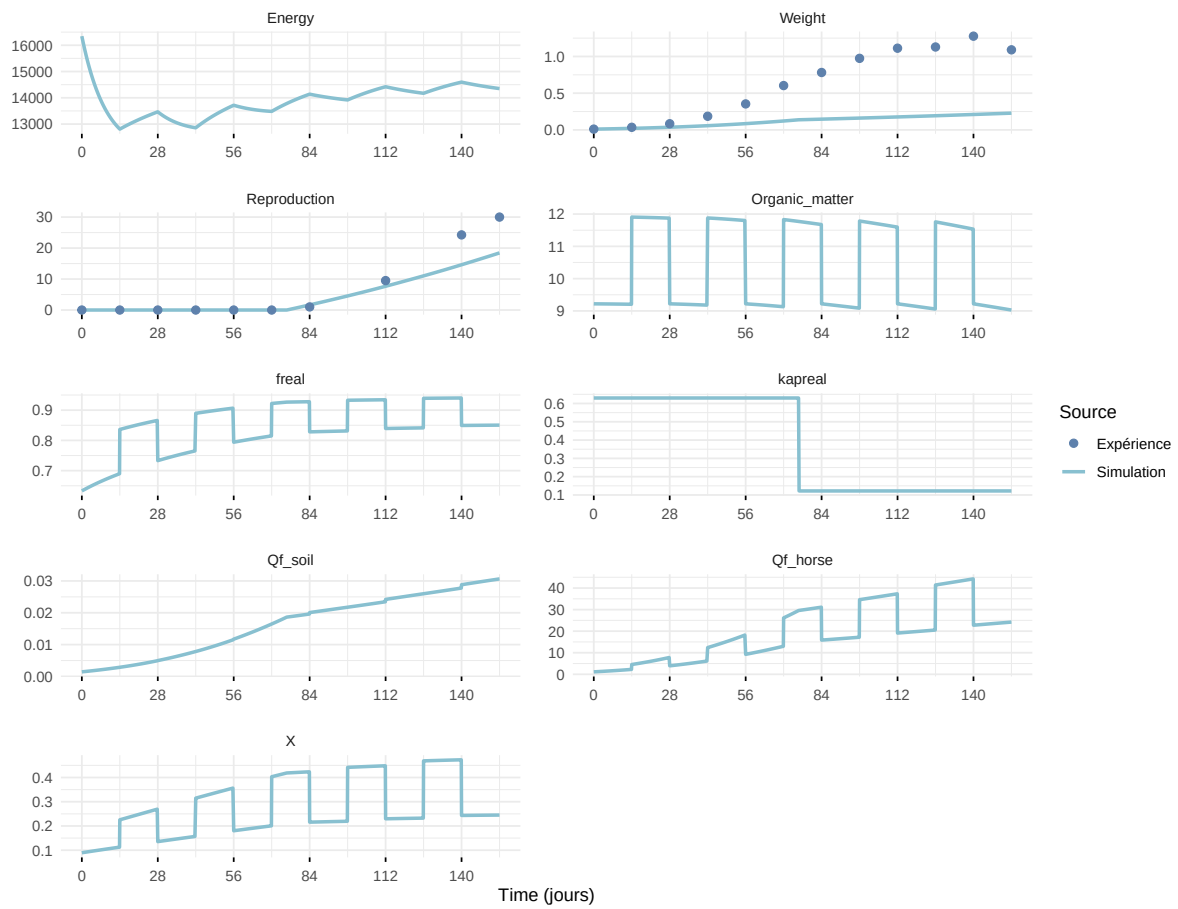
Expérience : Bart2020



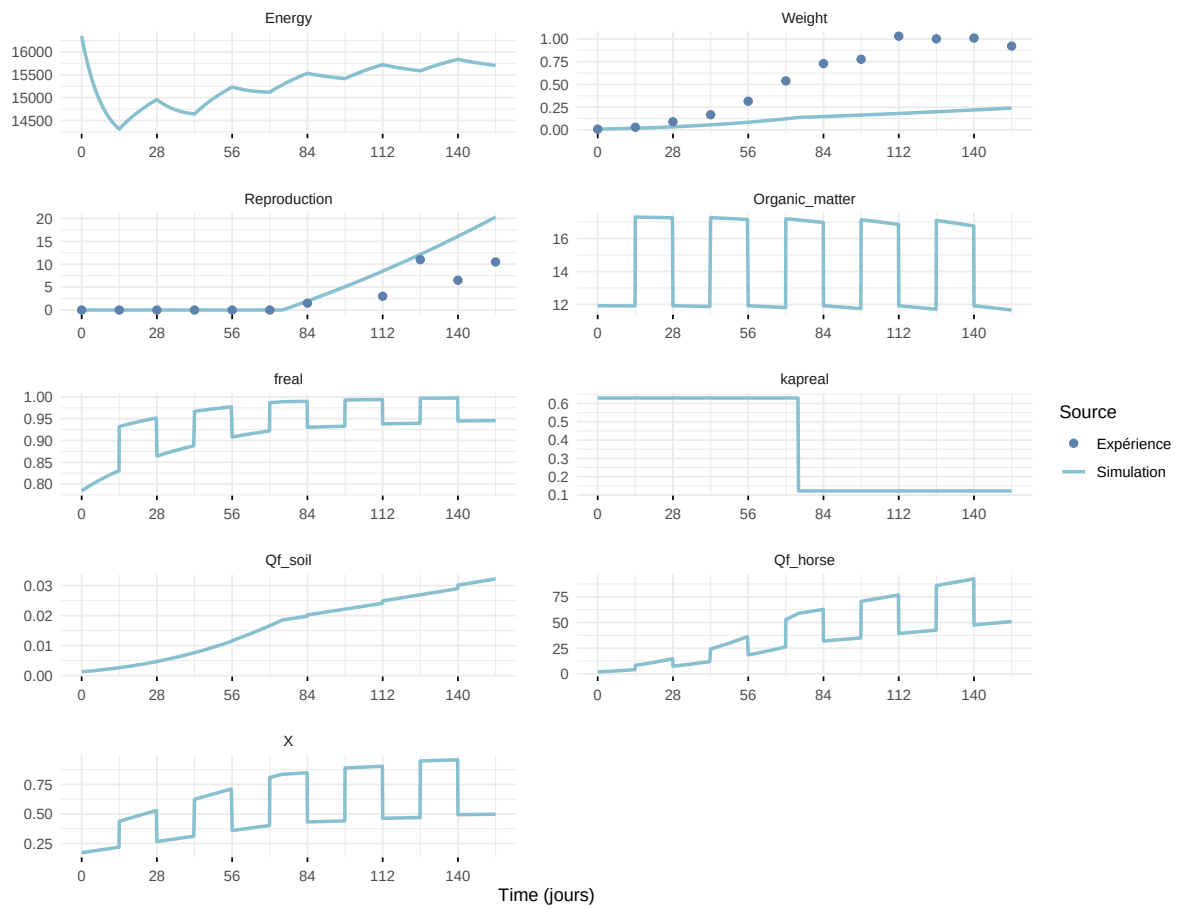
Expérience : Gollot2026_dens_D1



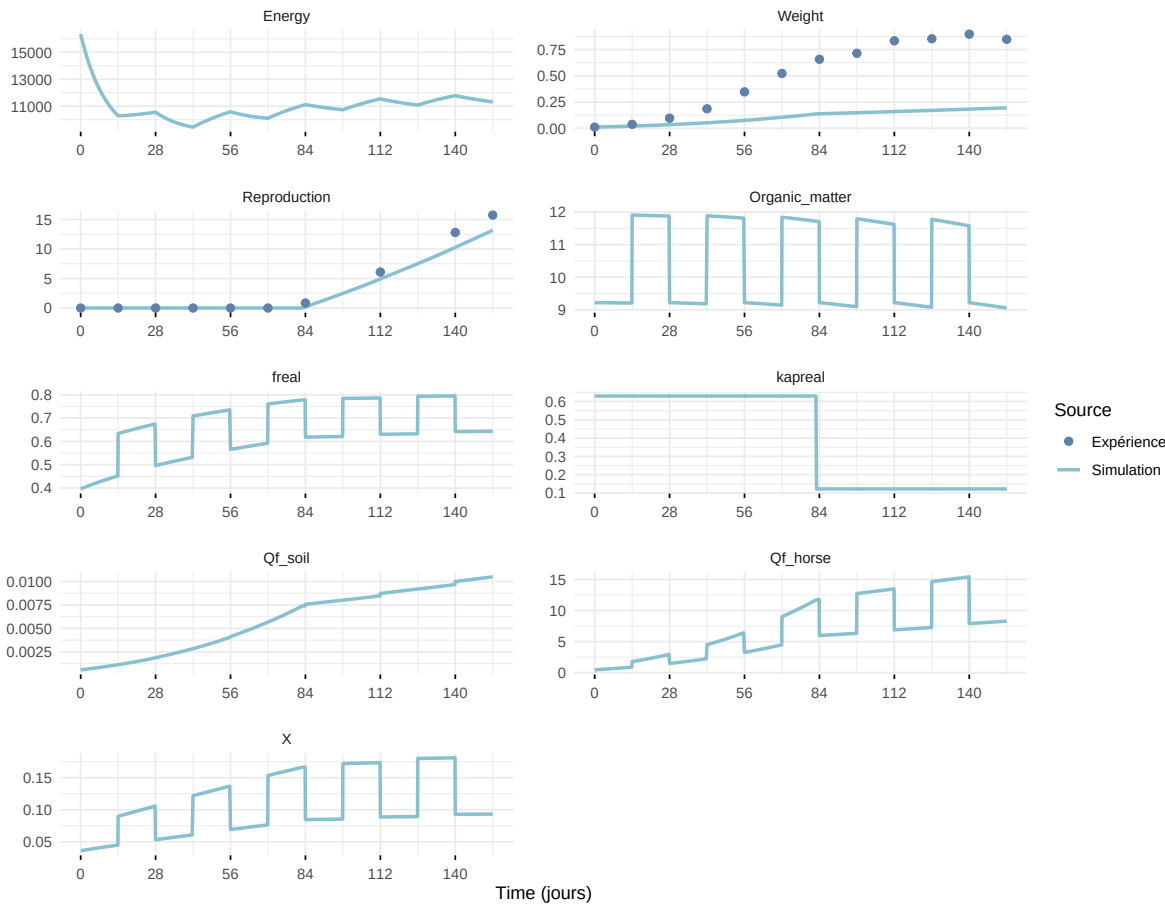
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

